Assignment Assessment (Group)

AMSE1003 Software Engineering

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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Diploma in Software Engineering

Programme:DSF (Group: 1)

Assignment

AMSE1003 SOFTWARE ENGINEERING

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Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted works are all our own work and are in our own words.

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Report for Club Membership Registration System

- Location: Tsung Tsin Secondary Middle School.
- Current operation method: Managing documents in printed forms and manual name list.
- Organization Background: Tshung Tsin Secondary Middle School manages a wide variety of student clubs, societies, and interest groups as part of its co-curricular program. Currently, administrative duties such as registering new members, handling club dues, tracking weekly attendance, and sending event notices are handled manually through physical paperwork and unstructured lists.

Part 1 : Introduction

Problems of Existing System

The problems of the manual process we faced:

- Inconvenient Attendance Tracking:

Attendance is currently recorded on loose paper sign-in sheets during club meetings and activities. These papers are easy to lose, take a lot of time to double-check manually, and make it very slow to calculate total attendance percentages or put together end-of-term reports for the school.

- Payment Bottlenecks & Manual Receipt Verification:

Since there is no central system for payments, the treasurer has to check every single paper receipt or bank transfer proof manually. This causes major delays in approving memberships, leads to payment record mistakes, and makes it hard to manage different fee tiers or late penalties properly.

- Missed Announcements:

Club updates and event details are usually pinned on physical noticeboards or lost in active group chats. Students often miss these announcements or forget past details because there is no single place to search for old notices, which leads to low turnout at events.

- Disorganized Record Management:

Student details are spread across physical logbooks and separate spreadsheets. Trying to find specific member records, make updates, or generate basic membership lists requires a lot of manual effort from advisors and committee members.

- Data Security & Privacy Risks:

Keeping personal information in physical files or unprotected spreadsheets leaves student details exposed. Without proper security measures or restricted access, unauthorized people could easily view sensitive data, creating privacy and security risks for students.

4 Major Software Quality Attributes Of The Proposed System

A good software should deliver the required functionality and performance to the user should be dependable, maintainable and usable. That would include :

1. Acceptability

- Acceptability in a Club Management System refers to the extent to which the system is accepted and used by club administrators and members. The system must be practical enough for club advisors to use during club activity to ensure a way faster experience. For example, the software could automatically organize students into their chosen clubs as soon as they sign up, rather than requiring the teacher to type each name into an Excel sheet by hand. At Tshung Tsin, it should directly reduce their workload by automating membership sorting and tracking rather than giving them more digital paperwork.

- The interface needs to be simple so club advisors can use it immediately without needing training sessions. It must also link smoothly with current school databases or excel sheets so the club committee can cross-reference student lists without doing manual data entry.

2. Dependability and security

- Dependability and security refers to a system's ability to remain reliable, perform correctly without failure, and protect its data from unauthorized access. In this system, this means maintaining a low failure rate so club advisors don't lose enrollment progress during peak registration periods, while simultaneously securing data privacy by encrypting sensitive student records—like contact details and IDs—to protect privacy and prevent data leaks. For example,the system can protect sensitive student data by hashing accounts' passwords using standard security protocols and encrypting personal ID numbers in the database.

- Only authorized advisors and designated committee members should be able to view, modify, or export student lists. This secures the database from unauthorized student access and potential tampering.

3. Efficiency

- Efficiency means a system's function to operate quickly and efficiently, even while under high load. In this project, it means the system processes club sign-ups and searches quickly so that advisors do not have to wait. Because many students will use the system at the same time during busy school hours, it must handle everyone without slowing down. This ensures that students may submit in their forms and committee members can approve them quickly. For example, during peak registration week at Tshung Tsin, hundreds of students may attempt to submit club application forms at the same time during recess.

- The system should allow committees to quickly search, update, and sort student lists without making the system slow or fail. By changing the old paper-based folders with an electronic digital system, the Tshung Tsin club management staff can complete their tasks more quickly, save computer memory, and reduce wait times.

4. Maintainability

- Maintainability is a system's ability to be easily changed, updated, repaired, and improved after it has been built. Maintainability is important in a Club Membership Registration System since club information, membership requirements, and registration procedures can all change over time. The system should be created with a clear and orderly structure so that developers may update club information, fix issues, and add new features without disrupting existing functionality. For example, if Tshung Tsin opens a new club or modifies the registration procedure, developers should be able to swiftly update the system without having to rebuild the app.
- The system should also have good documentation and a well-structured design, making it easier for future developers to understand and maintain. This minimizes maintenance time and costs while maintaining the system's reliability. As the number of clubs and members grows, the system can handle new requirements while maintaining daily registration and membership management.

Software Process Model

We recommend using Extreme Programming (XP) from the agile methodology. Our target is to develop our system within a 2 - 4 weeks period for this project system. We prefer this because the process is simple, easy , and most importantly, it’s to have group cohesiveness in the group. In addition, a kind senior whose experience in XP agrees to lend a hand in this project.

First, our leader and senior listens to the stories and feedback experienced by the club advisors and students. With the school stories and feedback on the current club membership, our leader analyzes and defines all the requirements of this system. Next, we create a simple and shared understanding system so the design provides clear direction and is manageable by our client users. After that, we developed a small release and tested the system to further estimate the future development more accurately. After we have tested the system, we can then let the client user test and provide any feedback regardless of good or bad. All of the developing, testing and feedback process is repeated until the client user approves it. Finally, we made the release of the approved system.

Part 2 : System Planning, Requirements and Architecture Design

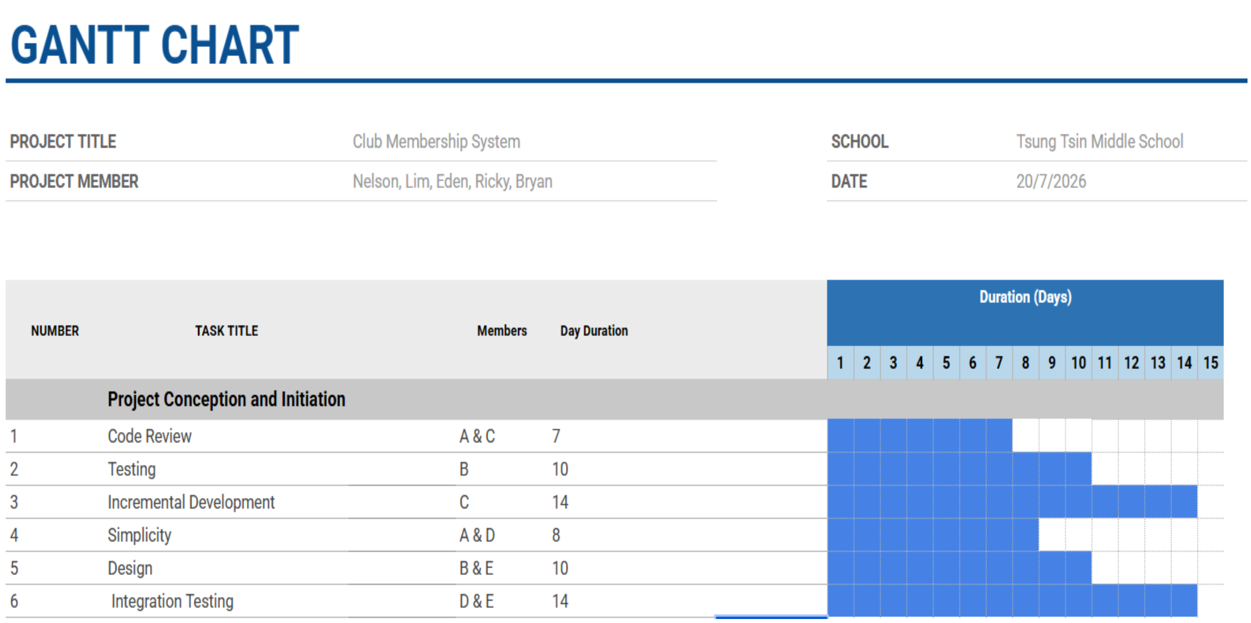
Project Plan and Schedule

Allocation List

Phase	Task Description	Assigned to…	Expected Duration
1. Code Review	Helps identify and fix errors quickly. XP promotes pair programming, where two developers work together and regularly switch roles.	Member A & C (Leader)	7 Days
2. Testing	Improves reliability by removing errors. XP follows Test-Driven Development (TDD), where tests are written before the actual code.	Member B	10 Days
3. Incremental Development	Software is built in small parts, allowing frequent feedback and continuous	Member C	14 Days

4. Simplicity	Keeping the design simple makes the code easier to develop, test, and maintain.	Member A & D	8 Days
5. Design	Continuous attention to design ensures better quality software.	Member B & E	10 Days
6. Integration Testing	Frequent integration and testing help detect issues early. XP supports continuous integration by testing multiple times a day.	Member D & E	14 Days

<u>Gantt Chart (Extreme Programming- XP)</u>



GANTT CHART

PROJECT TITLE	Club Membership System	SCHOOL	Tsung Tsin Middle School
PROJECT MEMBER	Nelson, Lim, Eden, Ricky, Bryan	DATE	20/7/2026

NUMBER	TASK TITLE	Members	Day Duration
	Project Conception and Initiation		
1	Code Review	A & C	7
2	Testing	B	10
3	Incremental Development	C	14
4	Simplicity	A & D	8
5	Design	B & E	10
6	Integration Testing	D & E	14

Software Requirements Specification

- **Treasurer module**

Description:Enables the club treasurer to manage membership fee payments, track dues, and maintain organized financial records.

Function Requirements:

1.1 The Treasurer module shall allow the treasurer to view pending registration payments, inspect attached payment receipts, and accept or reject applications to update account status.

1.2 The system shall allow the Treasurer to configure and update base registration fees, annual membership dues, and late penalty charges across distinct member tiers.

1.3 The system shall generate real-time revenue summary dashboards and allow the Treasurer to export financial statements as PDF files.

1.4 The system shall send automated or manual payment requests and renewal notices containing direct payment links to members via email or system notification.

Non-Function Requirements:

1.5The system shall encrypt all stored and transmitted financial data using standard encryption protocols.

- **Reporting module**

Description : The reporting module allows the club advisors to view,generate,and manage membership reports. It helps club management to monitor membership registration record efficiency.

Function Requirement:

2.1 The system enables users to generate a list of registered members.

2.2 The system allows advisors to view membership reports and save it in PDF or Excel format.

2.3 The system enables management to modify the data of membership registration list such as delete member records and add new members.

2.4 The system allows advisors to search reports that display the registration date and member name.

Non-functional Requirement:

2.5 The system shall ensure that only the authorized users such as advisor and committee members can access and view system information

- **Attendance module**

Description:The attendance module allows advisors and members to manage and view attendance records for club activities.

Functional requirements:

3.1 Advisors can record member’s attendance during club events and meetings.

3.2 Enable advisors to edit or remove the incorrect attendance records to keep the attendance information accurate.

3.3 Allow members to check their attendance status after each club activities.

3.4 The system allows registered members to view their attendance records and attendance percentage.

Non-functional requirements

3.5 The system should ensure that attendance records are correct and free from duplicate entries.

- **Announcement module**

Description: The Announcement module allows advisors to publish important club announcements and enables members to receive the latest updates about club activities and events.

Functional requirements

4.1 Allow advisors to create and publish announcements for club members.

4.2 Provide the latest club announcements and updated events to members.

4.3 Display the announcement history for the advisor to review previous notices.

4.4 Give members the ability to search announcements by title or publication date.

Non-functional requirements

4.5 Maintain fast loading of announcements,even when many users access the system simultaneously.

- **Registration Module**

The registration module helps new students join the club easily. It handles form submissions, profile setups, initial fee checks, and approval by club advisors.

Functional requirements :

5.1 Allow new students to submit an online form with their personal details.

5.2 Allow students to upload needed files (such as student card photos or payment receipts) when signing up.

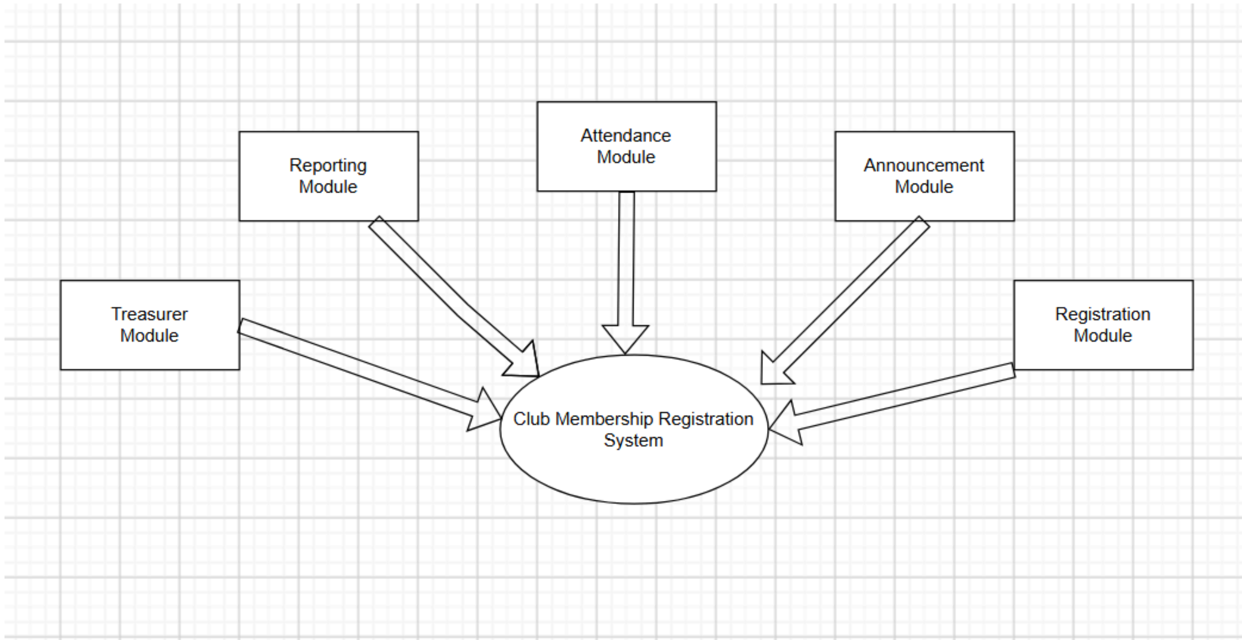
5.3 Allow club advisors to check new applications.

5.4 Send an automatic update message (by email or system alert) to the student when their application is accepted or declined.

Non-Functional Requirement:

5.5 The system must check all user input carefully to keep personal information safe during submission.

Architectural Design



We choose Repository Model, this is because Repository Model allows us to share data to one central database and each sub-system can maintain its own database and interchange data by passing messages. For example, data entered in one module is immediately available to others. Other than that, models are independent of one another so when modifying the user interface or logic of the Announcement Module will not break the Treasurer or Reporting modules. Last but not least, it is very easy to add new modules in the future, just by connecting them to the existing central database. In conclusion, the Repository Model works best here, because having one shared database keeps all student records accurate and in sync, so modules don't end up with conflicting information.

Part 3 : Software Testing and Configuration Management

Test Cases

Students Registration Module					
NO	Objective	Test Data	Expected result	Actual result	Remarks Comments
1	To submit an online form with their personal details using valid data.	Name, StudentID, Email address, Phone Number ,Class	New student personal details submitted.	-	-
2	To upload needed files when signing up.	Student card photo, Payment receipt	Required files are successfully uploaded .	-	-
3	Allow club advisors to view new applications.	New application record, Student details, Uploaded documents	Club advisors can view all new applications.	-	-
4	Send an automatic update message to the student when their application is accepted.	Application status, Student email address	Students receive an automatic message.	-	-

5	To allow students to check their application status online.	Student ID , Application status	Students can view their current application status successfully.	-	-

Report Module					
No	Objective	Test Data	Expected Result	Actual result	Remarks Comments
1.	To generate a list of registered members using valid data.	Type of club, Member ID,Registration Date,Phone number	A registered members list are generated successfully	-	-
2.	To view membership reports and save it in PDF or Excel format using valid data.	Member ID,Member Name,PDF/Excel format	Membership reports are saved in PDF/Excel successfully.	-	-
3.	To modify the data of membership registration list such as delete member records and add new members using valid data.	Member ID,Member Name,Registration Date,Add or Delete option	Member records are successfully modified	-	-
4.	To search reports that display the registration date and member name using valid data.	Member Name,registration Date	The system displays the matching member record.	-	-

5.	To ensure that only the authorized users such as advisor and committee members can view system information.	-	Only authorized users such as advisors and committee members can view system information.	-	-

Announcement Module					
NO	Objective	Test Data	Expected result	Actual result	Remarks/ Comments
1	To allow advisors to create an announcement	Date, text information	Advisors created an announcement successfully.	-	-
2	To allow advisors to publish an announcement	Announcement, event update, notices	Advisors publish an announcement successfully.	-	-
3	To allow members to view latest club announcement	Announcements, events, updates, notices	Members are able to view the latest announcements successfully.	-	-
4	To display the announcement history	Announcements, events, updates, notices	Advisors and members are able to see the display for announcement history successfully	-	-

5	To search for announcements using valid data	title , published date, author	Advisors and members are able to search for announcements successfully.	-	-

Attendance Module					
NO	Objective	Test Data	Expected result	Actual result	Remarks/ Comments
1	Allow advisors to record member's attendance during club events and meetings.	MemberID, EventID, Meeting Date, Attendance Status (Present/Absent)	Attendance record successfully added for the member.	-	-
2	Enable advisors to edit or remove the incorrect attendance records to keep the attendance information accurate.	Existing Attendance Record ID, Updated Status / Delete Request	Incorrect attendance record is successfully modified or deleted.	-	-
3	Incorrect attendance record is successfully modified or deleted.	MemberID, ActivityID	Members can view their attendance status for the selected activity.	-	-

4	The system should ensure that attendance records are correct and free from duplicate entries.	Duplicate Attendance Record (Same MemberID, Same EventID)	The system detects duplicate input and prevents the record from being created.	-	-

Treasurer Module					
NO	Objective	Test Data	Expected result	Actual result	Remarks Comments
1	To view pending registration payments using valid data.	Treasurer login credentials and a filter setting	Treasurer able to view pending registration payments.	-	-
2	To update base registration fees using valid data.	The fee name and new amount	Can update selected registration fees.	-	-
3	To generate real-time revenue summary dashboards as PDF files using valid data.	Date range and export format	Able to display real-time revenue summary dashboards as PDF files.	-	-
4	To send automated or manual payment requests to members via email or system notification using valid data.	Recipient details and message	Able to send automated or manual payment requests.	-	-
5	To encrypt all stored data.	Sample sensitive data to encrypt.	The system encrypts all stored data.	-	-

Software Configuration Management

Link to GitHub : https://github.com/IDGlitchMSchool/DSF-Assignments/blob/c2a1426c62ac3dcc8df5108e7195e3f58765c844/Software%20Engineering%20Assignment%20.pdf

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